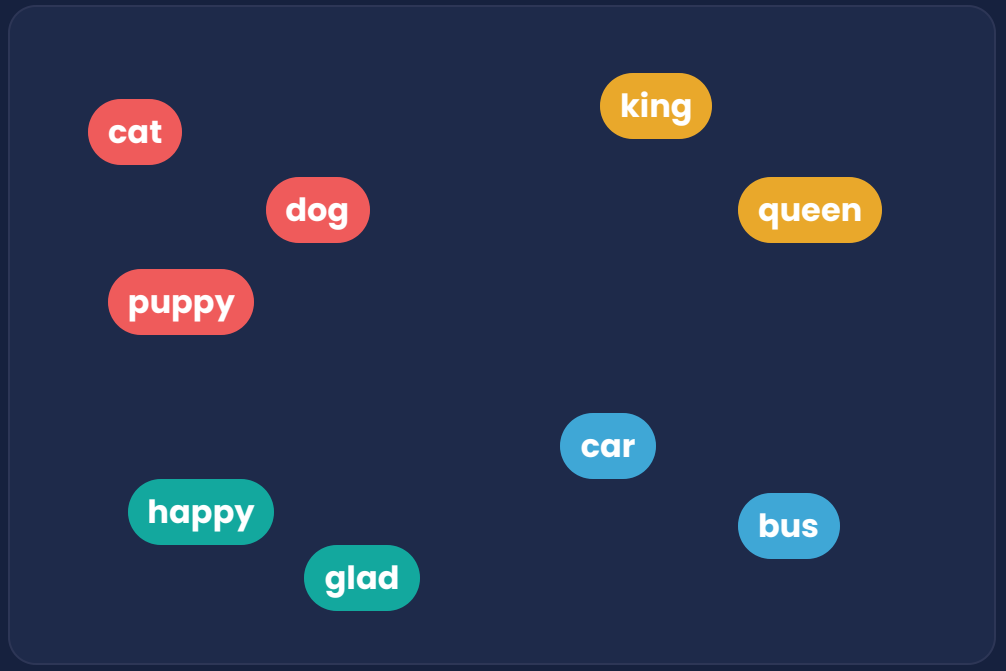
NATURAL LANGUAGE PROCESSING · LESSON 3

Teaching Computers What Words Mean

Word embeddings, similarity, word math, and building a smart search tool.



WARM-UP

Which word does not belong?

Look at these four words. Which one does not belong?

breakfast | lunch | dinner | school

For you, it is easy

You instantly know that three are meals and one is a place. You understand what the words **mean**.

For a computer, it is hard

To Bag of Words and TF-IDF, these are just four different columns. They have no idea that three of them are related.

By the end of today, your computer will answer this question correctly. Let’s find out how.

THE PROBLEM

Counting words is not understanding words

A shopper types **“cheap”** into an online store’s review search. Here is what TF-IDF finds.

Review	TF-IDF match score for “cheap”
The shoes were affordable and comfortable.	0.00
Great price, I saved a lot of money.	0.00
The food tasted delicious and fresh.	0.00
The screen cracked after one week.	0.00

Zero matches! The first two reviews are clearly about cheap things, but they never use the exact word “cheap.”

WHY THIS MATTERS

Computers need to understand meaning



Smarter search

Search engines find pages about “affordable laptops” even when you type “cheap computers.”



Recommendations

Apps suggest videos, songs, or articles that are similar in meaning to what you already like.



Translation

Translators match words with the same meaning across languages, like “happy” and “masaya.”



AI chatbots

Chatbots must understand that “How do I fix my phone?” and “My cellphone is broken” ask the same thing.

QUICK RECAP

From counting words to understanding them

What we learned in Lesson 2

Bag of Words and TF-IDF turn text into numbers by **counting** words. This was good enough to catch spam with 98.6% accuracy.

What they cannot do

Each word gets its own separate column. To the computer, “happy” and “glad” are complete strangers, just like “happy” and “table.”

Today’s big idea: give every word a **list of numbers that captures its meaning**. Words with similar meanings will get similar numbers.

LEARNING GOALS

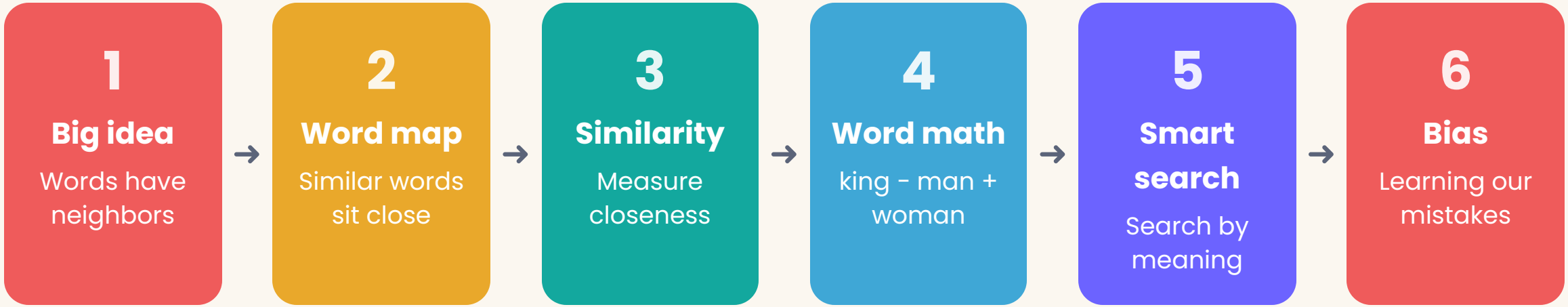
What we will learn today

1. Explain why counting words is not enough to understand meaning.
2. Describe how a computer learns meaning from the words around a word.
3. Measure how similar two words are using cosine similarity.
4. Solve word puzzles using “word math.”
5. Build a smart search tool that searches by meaning.
6. Explain how embeddings can learn unfair bias from people.

ROADMAP

Today's journey

Here is our plan for today's lesson.



01

The Big Idea

A word is known by the company it keeps.

THE BIG IDEA

Words are known by their neighbors

Fill in the blank. What words could go here?

"I drank a cup of hot ? this morning."

coffee | tea | chocolate | soup | bicycle

Words that fit in the **same blanks** usually have **similar meanings**. Coffee and tea are both drinks.

A linguist named J.R. Firth said it in 1957: "You shall know a word by the company it keeps."

THE BIG IDEA

How a computer learns meaning

Step 1: Read a LOT of text

The model we will use, called **GloVe**, read about **6 billion words** from Wikipedia and news articles.

Step 2: Notice the neighbors

It counted which words often appear near each other. “Coffee” often appears near “cup,” “hot,” and “drink.” So does “tea.”

Words that share many neighbors end up with **similar numbers**. No human had to label anything!

THE BIG IDEA

Each word becomes a list of numbers

A **word embedding** is a list of numbers that represents the meaning of a word. Our GloVe model gives each word **50 numbers**.

```
king → [0.50, 0.69, −0.60, −0.02, 0.60, −0.13, … 44 more]
```

The model knows **400,000 words**, and every one of them has its own list of 50 numbers.

Unlike Bag of Words, the list is **short** and has **no zeros wasted**. Every number carries a little bit of meaning.

THE BIG IDEA

A made-up example to build intuition

To understand the idea, imagine we described words with just three traits. These numbers are **made up** for learning.

Word	Is it alive?	Can you eat it?	Is it big?
cat	**0.9**	0.1	0.2
dog	**0.9**	0.1	0.3
pizza	0.0	**0.9**	0.2
bus	0.0	0.0	**0.9**

“Cat” and “dog” have almost the same numbers, so the computer knows they are similar.

Real embeddings work the same way, but the 50 traits are learned automatically and do not have simple names.

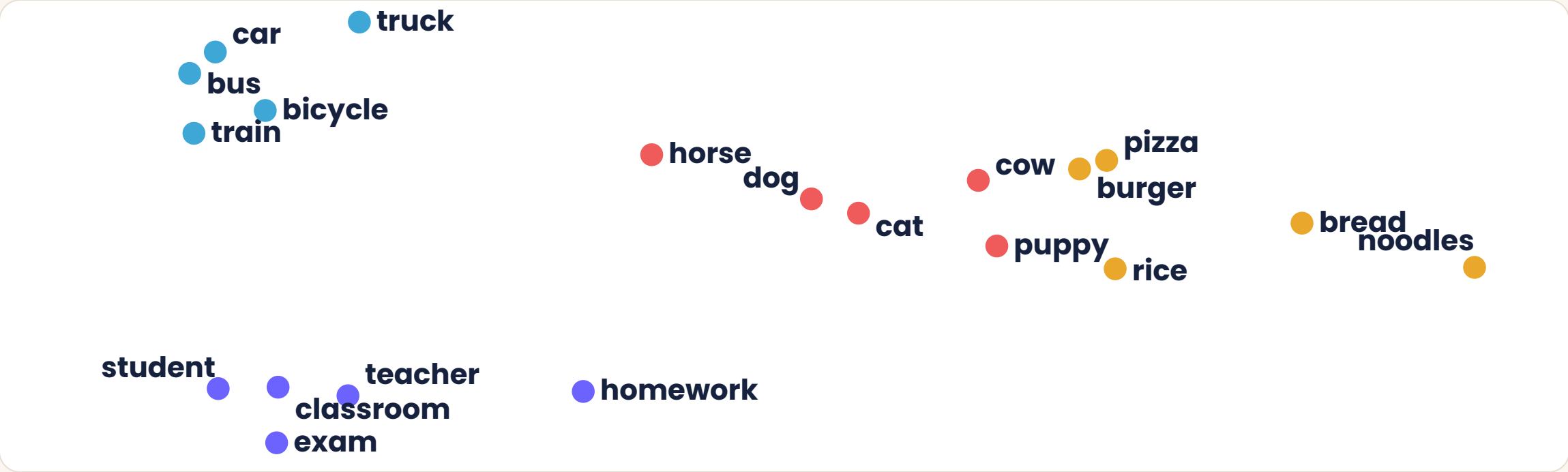
02

The Word Map

Similar words live in the same neighborhood.

SIMULATED DEMO

A real map of word meanings



These are **real GloVe embeddings**, squashed from 50 numbers down to 2 so we can draw them.

Nobody told the computer which words are vehicles or school words. It found the groups **by itself**.

Food sits near animals. Both often appear in texts about farms and eating.

Finding a word's neighbors

We can ask the model: "Which words are closest to this word on the map?"

Word	Its closest neighbors
pizza	`sandwich` `sandwiches` `snack` `bakery` `fries`
basketball	`football` `hockey` `baseball` `nba` `soccer`
school	`college` `schools` `campus` `graduate` `elementary`
smart	`intelligent` `sophisticated` `makes` `savvy` `easy`

Most neighbors make sense, even though the model only learned from reading text. A few, like "makes," are just words that often appear nearby.

SIMULATED DEMO

Exploring embeddings in Python

SIMULATED DEMO

```
import gensim.downloader as api

# Download 400,000 words with 50 numbers each
model = api.load("glove-wiki-gigaword-50")

print(model["king"][:6].round(2))

for word, score in model.most_similar("basketball",
topn=5):
    print(word, round(score, 2))
```

OUTPUT

```
[ 0.5   0.69 -0.6  -0.02  0.6  -0.13]
football 0.88
hockey 0.86
baseball 0.86
nba 0.84
soccer 0.82
```

gensim is a Python library for working with word embeddings.

model["king"] shows the list of numbers for a word. We print only the first 6.

most_similar() finds the nearest neighbors on the word map, with a closeness score.

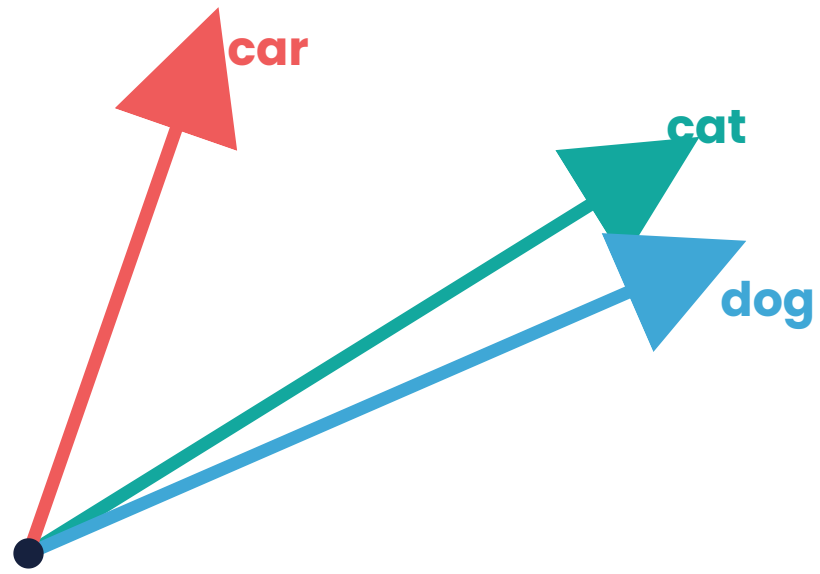
03

Measuring Similarity

How close are two words in meaning?

SIMILARITY

Cosine similarity: do the arrows point the same way?



Think of each embedding as an **arrow** pointing in some direction.

Cosine similarity checks how much two arrows point the **same way**. “Cat” and “dog” point almost the same way. “Car” points somewhere else.

The score goes up to **1**. A score near 1 means very similar. A low score means not related.

SIMILARITY

How similar are these words?

Here are real similarity scores from our GloVe model.



"Happy" and "glad" finally look similar to the computer, unlike in Bag of Words!

SIMULATED DEMO

Solving the warm-up puzzle

SIMULATED DEMO

```
print(round(model.similarity("cat", "dog"), 2))
print(round(model.similarity("cat", "car"), 2))

# Remember the warm-up puzzle?
print(model.doesnt_match(["breakfast", "lunch",
                          "dinner", "school"]))
```

OUTPUT

```
0.92
0.36
school
```

similarity() gives the cosine similarity of two words.

doesnt_match() finds the word that is farthest from the others.

The computer solved our warm-up puzzle! It picked **“school”** as the word that does not belong.

SURPRISE!

Opposites can look similar

happy & sad

0.69

Opposites, but the score is fairly high.

Closest word to “cheap”

expensive

Its score is 0.86, higher than “affordable” (0.71).

Why? Opposites appear in the **same kinds of sentences**: “The price is cheap” and “The price is expensive.” Embeddings measure how **related** words are, not whether they mean the same thing.

04

Word Math

Adding and subtracting meanings.

WORD MATH

The famous word equation

Because words are numbers, we can do math with them. Watch this famous example.



Start with “king.” It has the idea of royalty and the idea of male.

Subtract “man.” This removes the male part. Then **add “woman.”**

The closest word to the result is “queen,” with a score of 0.85.

WORD MATH

More word puzzles the computer can solve

Word math	The computer's answer	What it learned
`king − man + woman`	**queen**	Male and female versions
`paris − france + japan`	**tokyo**	Capital cities
`manila − philippines + japan`	**tokyo**	Capital cities, starting from Manila

The computer was never taught what a capital city is. It discovered the pattern from reading text.

SIMULATED DEMO

Word math in Python

SIMULATED DEMO

```
# positive = words to ADD
# negative = words to SUBTRACT
answers = model.most_similar(
    positive=["king", "woman"],
    negative=["man"],
    topn=3)

for word, score in answers:
    print(word, round(score, 2))
```

OUTPUT

```
queen 0.85
throne 0.77
prince 0.76
```

The words in **positive** are added together.

The words in **negative** are subtracted.

The answer is the nearest word to the result. “Queen” wins, and the next answers are also about royalty.

05

Smart Search

Searching by meaning, not exact words.

SMART SEARCH

From words to sentences

To compare whole sentences, we need one list of numbers per sentence. The simplest way is to **average** the word embeddings.

"The shoes were affordable and comfortable."

↓ remove stop words, look up each word

`shoes → [50 numbers]` `affordable → [50 numbers]` `comfortable → [50 numbers]`

↓ take the average

`sentence → [50 numbers]`

SIMULATED DEMO

Our smart search in action

We searched 10 product reviews. Here is the top result for each search.

Search	Top review found by meaning	Embedding score	TF-IDF score
cheap	The shoes were affordable and comfortable.	**0.76**	0.00
yummy meal	The burger was tasty but cold.	**0.74**	0.00
broken	The screen cracked after one week.	**0.75**	0.00
late shipping	The package arrived late and damaged.	**0.84**	0.46

Three of these searches share **no words** with the reviews, so TF-IDF scored 0.00. Embeddings still found the right answer.

SMART SEARCH

Not perfect yet

Let’s look at the full top three results for the search **“cheap.”**

Rank	Review	Score
1	The shoes were affordable and comfortable.	0.76
2	Great price, I saved a lot of money.	0.69
3	This phone is too **expensive** for students.	0.68

Result 3 is the opposite of what we wanted. This is the “opposites look similar” problem again.

EXPERIMENT

Do embeddings make a better spam detector?

Can embeddings improve the spam detector from Lesson 2? We tested it on the same 1,393 exam messages.

Text to numbers	Model	Accuracy	Spam missed
Bag of Words (Lesson 2)	Naive Bayes	**98.6%**	17
Averaged embeddings	Logistic Regression	95.3%	41

Averaging **blurs** strong clue words like “claim” and “150p” into one fuzzy meaning.

“Fr33” is not in the embedding’s vocabulary at all, so tricky spellings still fool it.

Lesson: a fancier method is not always better. Pick the tool that fits the task.

06

Bias and Limits

Embeddings learn our mistakes too.

BIAS

Embeddings can learn stereotypes

Embeddings learn from human writing, so they also learn human **stereotypes**.

Word math	The computer's answer
`doctor − man + woman`	**nurse**
`nurse − woman + man`	**doctor**

Women can be doctors and men can be nurses. The model learned an **unfair pattern** from old texts, not a fact.

BIAS

Why bias in AI matters



Job screening

If a hiring tool uses biased embeddings, it could rank people unfairly because of their gender.



Search results

Biased search could show only one kind of person for jobs like “engineer” or “nurse.”



What engineers do

They test models for bias, use more balanced data, and adjust embeddings to reduce unfair patterns.

An AI model is only as fair as the data it learns from. Testing for bias is part of an engineer’s job.

LIMITS

Whose language does the model know?

Our model learned mostly from English Wikipedia and news. Look what it thinks about Filipino words.

Word	Its closest neighbors
adobo	`chipotles` `habanero` `poblano` `chilies`
jeepney	`boeing-737` `open-top` `jeepneys` `cabs`

To this model, “adobo” is a Mexican chili sauce, not our favorite Filipino dish.

To understand Filipino meanings, a model must learn from **Filipino and Taglish text**.

LIMITS

Other limits of word embeddings

One meaning per word

“Bank” gets one list of numbers, whether it means a river bank or a money bank.

Unknown words

Words the model never read, like “fr33” or new slang, have no embedding at all.

Opposites look close

“Cheap” and “expensive” sit near each other because they appear in similar sentences.

Modern AI chatbots fix the first problem with **contextual embeddings**. That is our next big idea.

WHAT'S NEXT

From word embeddings to modern AI

Word embeddings (today)

Each word always gets the **same** list of numbers.

```
"river bank" → bank = [same]
"money bank" → bank = [same]
```

Contextual embeddings (modern AI)

Each word gets **different** numbers depending on the sentence around it.

```
"river bank" → bank = [nature]
"money bank" → bank = [money]
```

AI chatbots use a much bigger version of this idea, with models called **transformers**.

YOUR JOURNEY

The NLP journey so far

Look how far you have come in this NLP series.

L1
Clean text
Tokens, lemmas, stop words

→

L2
Count words
Bag of Words, TF-IDF

→

L3
Understand words
Embeddings

→

Next
Understand context
Transformers

Each lesson builds on the one before it. Modern chatbots use all of these ideas together.

SUMMARY

Key takeaways

1. Counting words cannot tell that “happy” and “glad” mean almost the same thing.
2. A word embedding is a list of numbers that captures a word’s meaning.
3. Computers learn meaning from the neighbors a word usually has.
4. Cosine similarity measures how close two words are in meaning.
5. Word math like king - man + woman ≈ queen shows that embeddings capture patterns.
6. Averaging word embeddings lets us search sentences by meaning.
7. Embeddings can learn unfair bias, so engineers must test them carefully.

CHECK YOUR UNDERSTANDING

Quick quiz

Q1

What is a word embedding?

A. A list of numbers that represents a word's meaning
B. A count of how many times a word appears
C. A dictionary definition

Q2

How does a model learn a word's meaning?

A. A teacher labels every word
B. It looks at the words that appear near it
C. It counts the letters

Q3

A cosine similarity near 1 means the words are...

A. Opposites
B. Very similar or related
C. Misspelled

Q4

What is king - man + woman?

A. prince
B. queen
C. man

Q5

Why did the model say doctor - man + woman = nurse?

A. It is a scientific fact
B. It learned a stereotype from human text
C. It made a spelling error

CHECK YOUR UNDERSTANDING

Quiz answers

Q1

Answer: A

Each word gets a list of numbers, like 50 for GloVe.

Q2

Answer: B

Words with similar neighbors get similar numbers.

Q3

Answer: B

The arrows point in almost the same direction.

Q4

Answer: B

The closest word to the result is “queen.”

Q5

Answer: B

Embeddings copy patterns from the text they read, including unfair ones.

ACTIVITY

Try it yourself

Activity 1: Human word map

Each group gets 12 word cards. Place them on the board so that similar words are close together. Then compare your map with the computer's.

Activity 2: Word math challenge

Invent three word equations, like "puppy - dog + cat." Predict the answer, then test it in the notebook.

Activity 3: Bias detective

Test word pairs for jobs and genders. Write down any unfair patterns you find and explain where they came from.

Bonus: Which Filipino words would you use to test if a model really understands our language?

NATURAL LANGUAGE PROCESSING

Thank you for listening!

Today you taught a computer what words mean.
What questions do you have?

`king` `−` `man` `+` `woman` `≈ queen`